

BRIAN MUSUMBA MWARO

 +254 726 045 736

 brianmwaro3@gmail.com

 Nairobi, Kenya

PROFESSIONAL SUMMARY

Results-driven Biomedical Engineering professional with over 3 years of progressive experience in installation, preventive maintenance, calibration, and repair of biomedical equipment within hospital environments. Proven expertise across ICU, theatre, renal, radiology, and laboratory systems.

Skilled in operation and maintenance of oxygen PSA plants, vacuum systems, compressors, and reverse osmosis (RO) systems. Strong working knowledge of ISO 13485, ISO 14971, and IEC 60601 standards, ensuring compliance, patient safety, and equipment reliability.

CORE COMPETENCIES

- Biomedical Equipment Installation, Maintenance & Repair
- Preventive Maintenance (PPM Planning & Execution)
- Calibration & Performance Verification
- ICU, Theatre, Renal, Radiology & Lab Equipment
- Oxygen PSA Plants, Vacuum & Compressor Systems
- Reverse Osmosis (RO) & Dialysis Systems
- Fault Diagnosis & Technical Troubleshooting
- ISO 13485, ISO 14971 & IEC 60601 Compliance
- Equipment Validation & Quality Assurance
- Maintenance Documentation & Reporting
- Health, Safety & Environmental Compliance
- Strong Analytical, Communication & Teamwork Skills

PROFESSIONAL EXPERIENCE

Biomedical Engineering Volunteer

Bungoma County Hospital

 March 2024 – October 2024

- Installed, inspected, and maintained biomedical equipment across ICU, theatre, renal, radiology, and laboratory units.
- Executed preventive maintenance schedules (PPM), reducing equipment downtime and improving service reliability.
- Serviced critical equipment including ventilators, dialysis machines, patient monitors, ECG machines, infusion pumps, and anesthesia machines.
- Conducted routine checks on oxygen PSA plants, compressors, manifolds, and vacuum systems to ensure continuous supply.
- Performed equipment calibration and verification using biomedical analyzers (e.g., Fluke ProSim 8).
- Supported maintenance of Reverse Osmosis (RO) systems used in dialysis.
- Maintained accurate documentation including job cards, service reports, and maintenance logs.
- Ensured full compliance with ISO 13485, ISO 14971, and IEC 60601 standards.
- Responded promptly to breakdowns, minimizing disruption to patient care.

Biomedical Engineering Attaché

Bungoma County Hospital



December 2021 – July 2023

- Installed, serviced, and repaired biomedical equipment in ICU, theatre, renal, and radiology departments.
- Conducted preventive maintenance and troubleshooting for patient monitors, CT scanners, and lab analyzers.
- Assisted in monitoring and maintenance of oxygen supply and vacuum systems.
- Supported maintenance of hospital infrastructure including electrical, HVAC, and plumbing systems.
- Performed basic calibration and electrical safety checks on medical devices.
- Maintained detailed maintenance records and ensured compliance with operational standards.

Biomedical Engineering Attaché

Kakamega County Hospital



March 2021 – April 2021

- Assisted in installation, maintenance, and repair of biomedical equipment.
- Conducted inspection and troubleshooting of electrical and electronic medical devices.

- Supported maintenance documentation and reporting.
- Assisted in electrical wiring installations and system upgrades within hospital departments.

Cyber Expert / IT Support

Humanitas Educational Centre



March 2025 – November 2025

- Provided technical support for computers, printers, and network systems.
- Diagnosed and resolved hardware and software issues, ensuring smooth operations.
- Managed data entry, document processing, and ICT system support.
- Maintained logs of technical issues and resolutions for accountability and tracking.
- Strengthened analytical and troubleshooting skills applicable to biomedical systems.

EDUCATION

1. Diploma in Biomedical Engineering

Kenya Medical Training College (KMTC) – 2024

2. Certificate in Medical Engineering

Kenya Medical Training College (KMTC) – 2022

3. Kenya Certificate of Secondary Education (KCSE)

St. Mathias Kholera Boys High School – 2018

CERTIFICATIONS & TRAINING

- Ajira Digital Program – 2024
- Data Analysis using Advanced Excel – 2024
- Training in Installation, Maintenance & Calibration of Medical Equipment

PROFESSIONAL ATTRIBUTES

- High level of integrity and accountability
- Strong attention to detail
- Ability to work under minimal supervision
- Excellent problem-solving skills
- Team-oriented with effective communication

REFEREES

1. Mr. James Kebaso

Head of Medical Engineering

KMTC – Kilifi Campus

Tel: +254 712 966 697

2. Rev. Tom Ndambo

Lead Pastor

The Sanctuary of Praise – Kilifi

Tel: +254 720 048 561

3. Eng. Masolo Samuel

Senior Biomedical Engineer

Bungoma County Hospital

Tel: +254 725 323 723